

Multiple Choice (10 marks)

1. The polynomial $x^4 + 4$ can be factored into linear factors in $\mathbb{Z}_5[x]$. Find this factorization.
 - (a) $(x - 2)(x + 2)(x - 1)(x + 1)$
 - (b) $(x+1)^4$
 - (c) $(x-1)(x+1)^3$
 - (d) $(x-1)^3(x+1)$
2. How many structurally distinct Abelian groups have order 72?
 - (a) 4
 - (b) 6
 - (c) 8
 - (d) 9
3. A tree is a connected graph with no cycles. How many nonisomorphic trees with 5 vertices exist?
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
4. If a real number x is chosen at random in the interval $[0, 3]$ and a real number y is chosen at random in the interval $[0, 4]$, what is the probability that $x < y$?
 - (a) $1/2$
 - (b) $7/12$
 - (c) $5/8$
 - (d) $2/3$
5. In xyz -space, what are the coordinates of the point on the plane $2x + y + 3z = 3$ that is closest to the origin?
 - (a) $(0, 0, 1)$
 - (b) $(3/7, 3/14, 9/14)$
 - (c) $(7/15, 8/15, 1/15)$
 - (d) $(5/6, 1/3, 1/3)$

True / False (10 marks)

Answer True or False

6. True or False: If H is a subgroup of G , then $aH = Ha$ holds only if H is a normal subgroup of G .
7. True or False: $(g \circ h)^2 = g^2 \circ h^2$ for every g, h in an abelian group G .
8. True or False: The only coset of the subgroup $4\mathbb{Z}$ in $2\mathbb{Z}$ is $2\mathbb{Z}$.
9. True or False: The value of $f(8)$ is 30, based on the recursive definition of the function $f(x)$.
10. True or False: There are exactly two abelian groups of order 4 up to isomorphism.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In right triangle ABC with $\angle B = 90^\circ$, we have $\sin A = 2 \cos A$. What is $\tan A$?
12. Discuss the implications of an eigenvector v of an invertible matrix A . Analyze whether v remains an eigenvector for the matrices $2A$, A^2 , and A^{-1} , providing justification for your conclusions.

End of Paper